Wally Ahmed

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SUMMARY

Rising software engineer and AI researcher with experience building **Agentic automation tools**, conducting applied **AI research**, and **developing data-driven solutions** for sustainability challenges. Experienced working on interdisciplinary research through The Grove School of Engineering at CCNY. **Building agentic workflows,** implementing **RAG systems,** creating **voice agents, and data analysis and research** to solve real-world problems. Passionate about artificial intelligence, software engineering, and using technology to create scalable environmental and societal impact. Seeking internships where I can contribute technical skills while continuing to grow as a software engineer.

TECHNICAL EXPERIENCE

Founder & Lead Engineer *OpenQuant-Agentic Trading Platform* **May 2026 – Present**

- Architected a **multi-tenant agentic trading platform** (**TypeScript, Node.js, PostgreSQL, React**) where autonomous AI agents research live markets, plan, and execute trades through a gated **research → plan → execute → review** loop against real brokerage APIs — **paper trading, options support**, and a full **KYC → account → funding** onboarding flow.
- Designed a **model-agnostic agent harness** spanning commercial LLM gateways (**300+ models**) and **self-hosted Hugging Face models with LoRA adapters** on serverless **GPU** infrastructure, letting users bring their own fine-tuned trading models.
- Built **defense-in-depth safety** into the core: **prompt-injection fencing**, a **deny-by-default tool gateway**, **fail-closed trade guards**, strict **per-agent tenant isolation**, and kill-switches — untrusted market data is neutralized before it can reach a model or a trade path.
- Engineered a **per-agent semantic memory layer** (every research cycle's lessons recalled via **semantic search**) and a **skills system** where agents author, version, and effectiveness-track their own trading strategies behind human and automated **approval gates**.
- Ships with **4,800+ automated tests**, **Stripe** billing, and a **React** dashboard with live portfolios and agent-memory search.

Technical Intern *Engineers Without Borders USA (CCNY Chapter)* **November 2025 – May 2026**

- **Developed Agentic NO-CODE pipelines** to curate reports and stories into interactive generated blog pages curated from **mutli-modal data interpretation by** leveraging **vision-models** in order to build the **UI/UX** for new blog pages.
- Migrated the chapter codebase from a static HTML to a fully interactive REACT/Next.js platform leveraging modern technologies to improve maintainability and adhere to best practices.
- Utilized AI development tools including **Claude, Cursor, ChatGPT-CodeX, GitHub Copilot, and Microsoft Copilot** to accelerate debugging, prompt engineering, and feature development.
- Owned ongoing site updates, maintenance, and deployments to keep content and functionality current.

Software Engineering Fellow *Springboard - Software Engineering Associate Track* **May 2023 – Jan 2025**

- **Over 800+ hours** of hands of completed hands on course-work with mentors developing fundamentals of **Frontend, Backend, Fullstack development, Database Management, and Software Architecture.**
- Under the mentorship of industry experts, I've engaged in hands-on learning, allowing me to build a strong foundation in both front-end (JavaScript, HTML, CSS, jQuery, Bootstrap) and back-end (Python, Flask, Node.js, Express, SQL, PostgreSQL) web development. My proficiency extends to RESTful APIs, Git/GitHub, data structures, Problem-Solving and Algorithm Design
- 1-year intensive boot-camp in Software Engineering methodologies and technologies.

LEADERSHIP EXPERIENCE

Hermes Handoff | Hackathon Winner **July 2026**

- **Placed 1st in the Cotal.ai sponsor track** at the **AGI Summit 2026 Hackathon** for building a wizard CLI that detects which AI coding harnesses are installed and signed in on a machine, then routes any task — research, files, data, or code — through the user's existing subscription instead of paid API credits.
- Built **multi-agent mesh connectors** in **TypeScript** for **OpenAI Codex CLI** and **Google Antigravity**, letting agents on different vendors' harnesses accept delegated work as autonomous workers.
- Designed a **12-task benchmark suite** (coding, agentic, recall, reasoning, hallucination resistance) and published **11 hermetic runs as 9 studies**, showing harness-per-model routing beat single-loop multi-model delegation by **~30% wall-clock** at equal quality.
- Proved fault tolerance live by killing a worker mid-task and measuring the mesh detect the failure, reassign the work, and finish it in **122 seconds**; documented the system as a **narrated 32-scene interactive walkthrough** deployed on **Fly.io**.

Live site: https://hermes-handoff-overview.fly.dev/

Flightdeck-AI Software Factory | Hackathon Winner **June 2026**

- **Placed 1st in the Render Track**, leading a team to build an AI-powered software delivery workflow with automated planning, coding, testing, review, and deployment stages.
- Directed development of a gated pipeline that generated preview deployments, allowing reviewers to test features before approving pull requests.
- Owned the technical stack — Render deployment services, AI-agent workflows, React/Vite dashboards, and persistent run-state tracking to support end-to-end software automation.

Live site: https://flightdeck-board.onrender.com/architecture

City Hacks The State | Hackathon Winner AI Facilities Intake & Work-Order Platform **June 2026**

- Placed **Top 5 overall** at the hackathon and won the **Best Architecture** award, leading the team that designed a scalable, AI-powered facilities-reporting system that accepts requests through web chat, Telegram, and voice calls.
- Built an agentic workflow to verify facility locations, classify issue priority, draft clear maintenance reports, and require user confirmation before submitting work orders.
- Connected AI intake workflows with a work-order dashboard, **GraphQL** APIs, **MongoDB**, **Telegram**, **Twilio**, and server-side routes that protected sensitive credentials.
- Coordinated the team across intake, backend, and dashboard workstreams, integrating each piece into one demo that students could use to submit building issues end to end.

Live site: https://ca-intake-ae320e.fly.dev/architecture.html

Open Source Contributor **Ongoing**

- **Authored four MIT-licensed open-source MCP (Model Context Protocol) projects** — three **TypeScript** servers plus a one-command installer — that let any AI coding agent delegate work to external models as subagents: **OpenRouter** (any model, with unified reasoning-effort and token-budget control), **OpenAI GPT** via both ChatGPT-subscription OAuth and API key, and **Google Antigravity** wiring.
- Shipped a reusable **orchestration-pattern library** inside each server so an agent retrieves a proven delegation workflow instead of improvising prompts; these servers were the subagent baselines in the Hermes Handoff benchmark study.
- **Contributed three upstream pull requests** to Cotal.ai's open-source agent-orchestration monorepo, including the **first Google Antigravity connector published anywhere** and a connector for OpenAI's Codex CLI, each ported from a working prototype to the maintainers' **TypeScript** standards.
- **Diagnosed and fixed a production defect** in which crashed agents stayed listed as online indefinitely — shipped five repairs in a single reviewed commit (12 files, +687/−9) with three new regression tests covering the failure modes.
- Kept the contribution merge-clean across four upstream releases, passing the project's full **typecheck, build, and CI smoke-test gates** on every rebase.

Link: https://github.com/Cotal-AI/Cotal/pulls?q=author%3AWally-Ahmed

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Bash/Shell
Frontend: React, Next.js, Vite, Tailwind CSS, HTML/CSS
Backend / Data: Node.js, Express, PostgreSQL, MongoDB, GraphQL, REST APIs, Docker
AI / Agents: LLM agent orchestration (multi-agent meshes, failover), MCP (Model Context Protocol) server development, agent benchmarking & evaluation, RAG, vector search (Pinecone), prompt-injection defense, Hugging Face fine-tuning (LoRA), voice agents (TTS/STT)
Cloud / DevOps: AWS, Microsoft Azure, Fly.io, Render, RunPod (serverless GPU), Stripe, Twilio, Git/GitHub, GitHub Actions, CI/CD, pnpm monorepos

CERTIFICATIONS

Google AI Essentials – Google **May 2026**
Digital Marketing Certified – HubSpot Academy **May 2026**

EDUCATION

Grove School of Engineering (CUNY) – Bachelor of Science (BS) in **Computer Science**